MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – WEEKLY CHAPTER TEST – 04** **Time:** 45 Minutes

Class: Class X (CBSE) **Chapter 4: Quadratic Equations** **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

Section A: MCQ & Assertion-Reason Questions

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Q1. If the equation $ax^2 + 2bx + c = 0$ has equal roots, then c is equal to: [1]

(A) $\frac{b^2}{a}$ **(B)** $\frac{b^2}{4a}$

(C) $\frac{a^2}{b}$ **(D)** $\frac{a}{b^2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\frac{b^2}{a}$

Comparing $ax^2 + 2bx + c = 0$ $(a \neq 0)$ with standard quadratic form $Ax^2 + Bx + C = 0$, we have $A = a$, $B = 2b$, and $C = c$.

Formula Used: Equal real roots $\iff D = B^2 - 4AC = 0$

Setting $D = 0$:

$$(2b)^2 - 4(a)(c) = 0 \implies 4b^2 = 4ac \implies \boxed{c = \frac{b^2}{a}}$$

Therefore, option **(A)** is correct. **[1 mark]**

Q2. If the quadratic equation $9x^2 + 6kx + 4 = 0$ has equal roots, then both roots are equal to: [1]

(A) $\pm\frac{2}{3}$ **(B)** $\pm\frac{3}{2}$

(C) 0 **(D)** ± 3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\pm\frac{2}{3}$

For $9x^2 + 6kx + 4 = 0$, $a = 9$, $b = 6k$, and $c = 4$.

Formula Used: $D = b^2 - 4ac = 0 \implies$ equal root $x = -\frac{b}{2a}$

Setting $D = 0$:

$$(6k)^2 - 4(9)(4) = 0 \implies 36k^2 = 144 \implies k^2 = 4 \implies k = \pm 2$$

Thus the double root is $x = -\frac{6k}{2(9)} = -\frac{k}{3} = \mp\frac{2}{3}$, i.e., $\boxed{\pm\frac{2}{3}}$.

Hence, option **(A)** is correct. **[1 mark]**

Q3. If the equation $ax^2 + bx + c = 0$ $(a \neq 0)$ has equal roots, then c equals: [1]

(A) $-\dfrac{b}{2a}$ **(B)** $\dfrac{b}{2a}$

(C) $-\dfrac{b^2}{4a}$ **(D)** $\dfrac{b^2}{4a}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D) $\dfrac{b^2}{4a}$

For $ax^2 + bx + c = 0$ $(a \neq 0)$ to have equal real roots, its discriminant must be zero:

Formula Used: $D = b^2 - 4ac = 0$

Solving for c:

$$b^2 - 4ac = 0 \implies 4ac = b^2 \implies \boxed{c = \frac{b^2}{4a}}$$

Therefore, option **(D)** is correct. **[1 mark]**

Q4. The positive value of k for which both the equations $x^2 + kx + 64 = 0$ and $x^2 - 8x + k = 0$ will have real roots is: [1]

(A) 4 **(B)** 8

(C) 12 **(D)** 16

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D) 16

We know that a quadratic equation has real roots if and only if its discriminant $D \geq 0$.

Formula Used: Real roots $\iff D = b^2 - 4ac \geq 0$

(a) For $x^2 + kx + 64 = 0$: $D_1 = k^2 - 4(1)(64) \geq 0 \implies k^2 \geq 256$. Since $k > 0$, we get $k \geq 16$.

(b) For $x^2 - 8x + k = 0$: $D_2 = (-8)^2 - 4(1)(k) \geq 0 \implies 64 - 4k \geq 0 \implies k \leq 16$.

The only positive value satisfying both $k \geq 16$ and $k \leq 16$ simultaneously is $\boxed{k = 16}$.

Hence, option **(D)** is correct. **[1 mark]**

Q5. Assertion-Reason: [1]

Assertion (A): If the roots of $(2k - 1)x^2 + 4x - 3 = 0$ are reciprocal to each other, then $k = -1$.

Reason (R): If the roots of a quadratic equation $ax^2 + bx + c = 0$ $(a \neq 0)$ are reciprocals of each other, then $a = c$.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true, but R is not the correct explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

Let the roots of $ax^2 + bx + c = 0$ $(a \neq 0)$ be α and $\frac{1}{\alpha}$.

Formula Used: Reciprocal roots $\implies \alpha \cdot \dfrac{1}{\alpha} = \dfrac{c}{a} = 1 \implies a = c$

Applying this to $(2k - 1)x^2 + 4x - 3 = 0$ where $a = 2k - 1$ and $c = -3$:

$$2k - 1 = -3 \implies 2k = -2 \implies k = -1$$

At $k = -1$, leading coefficient $a = 2(-1) - 1 = -3 \neq 0$, so the equation remains valid quadratic. Thus Assertion (A) is **true**, Reason (R) is **true**, and Reason (R) correctly explains Assertion (A). **[1 mark]**

Q6. Find the value of k for which the quadratic equation $kx(x-2)+6=0$ has two equal roots. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

First, expand and rewrite $kx(x-2)+6=0$ in standard form:

$$kx^2-2kx+6=0 \quad [\tfrac{1}{2}\ \textbf{mark}]$$

For this to be a quadratic equation, $k \neq 0$. Comparing with $ax^2+bx+c=0$, we have $a=k$, $b=-2k$, and $c=6$.

Formula Used: Two equal real roots $\iff D=b^2-4ac=0$

Setting $D=0$:

$$(-2k)^2-4(k)(6)=0 \implies 4k^2-24k=0 \implies 4k(k-6)=0 \quad \textbf{[1 mark]}$$

Therefore, $k=0$ or $k=6$.
If $k=0$, the equation becomes $6=0$ (contradiction; not quadratic), so we reject $k=0$.
Hence, the required value is $\boxed{k=6}$. $[\tfrac{1}{2}\ \textbf{mark}]$

Q7. Find the value(s) of k for which the quadratic equation $(k-12)x^2+2(k-12)x+2=0$ has real and equal roots. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given equation: $(k-12)x^2+2(k-12)x+2=0$.
Since it is a quadratic equation, leading coefficient $k-12 \neq 0$, i.e., $k \neq 12$.
Comparing with $ax^2+bx+c=0$: $a=k-12$, $b=2(k-12)$, and $c=2$. $[\tfrac{1}{2}\ \textbf{mark}]$

Formula Used: Real and equal roots $\iff D=b^2-4ac=0$

Computing and equating D to zero:

$$\begin{aligned}
[2(k-12)]^2-4(k-12)(2)&=0\\
4(k-12)^2-8(k-12)&=0 \quad \textbf{[1 mark]}\\
4(k-12)[(k-12)-2]&=0\\
4(k-12)(k-14)=0 &\implies k=12 \text{ or } k=14 \quad \textbf{[1 mark]}
\end{aligned}$$

We reject $k=12$ because substituting $k=12$ makes $a=0$ and reduces the equation to $2=0$ (inconsistent).
Hence, the only valid value is $\boxed{k=14}$. $[\tfrac{1}{2}\ \textbf{mark}]$

Q8. If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are real and equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Comparing with $Ax^2 + Bx + C = 0$, we identify $A = c^2 - ab$, $B = -2(a^2 - bc)$, and $C = b^2 - ac$.

Formula Used: Real and equal roots $\iff D = B^2 - 4AC = 0$

Setting $D = 0$:

$$\begin{aligned}\left[-2(a^2 - bc)\right]^2 - 4(c^2 - ab)(b^2 - ac) &= 0\\ 4(a^2 - bc)^2 - 4(c^2 - ab)(b^2 - ac) &= 0 \quad \textbf{[1 mark]}\end{aligned}$$

Dividing throughout by 4 and expanding both products:

$$\begin{aligned}(a^4 - 2a^2bc + b^2c^2) - (b^2c^2 - ac^3 - ab^3 + a^2bc) &= 0\\ a^4 - 2a^2bc + b^2c^2 - b^2c^2 + ac^3 + ab^3 - a^2bc &= 0 \quad \textbf{[1 mark]}\\ a^4 + ab^3 + ac^3 - 3a^2bc &= 0\\ a(a^3 + b^3 + c^3 - 3abc) &= 0\end{aligned}$$

Equating each factor to zero gives either $a = 0$ or $a^3 + b^3 + c^3 - 3abc = 0$, i.e.:

$$\boxed{a = 0 \quad \text{or} \quad a^3 + b^3 + c^3 = 3abc} \qquad \text{(Hence Proved)} \quad \textbf{[1 mark]}$$

Q9. Find the values of p for which the quadratic equation $(2p + 1)x^2 - (7p + 2)x + (7p - 3) = 0$ has equal roots. Also, find these roots. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Comparing with $ax^2 + bx + c = 0$, we have $a = 2p + 1$ $(p \neq -\frac{1}{2})$, $b = -(7p + 2)$, and $c = 7p - 3$.

Formula Used: Equal real roots $\iff D = b^2 - 4ac = 0$, Double root $= -\dfrac{b}{2a}$

Setting $D = 0$:

$$\begin{aligned}\left[-(7p + 2)\right]^2 - 4(2p + 1)(7p - 3) &= 0\\ (49p^2 + 28p + 4) - 4(14p^2 + p - 3) &= 0\\ 49p^2 + 28p + 4 - 56p^2 - 4p + 12 &= 0\\ -7p^2 + 24p + 16 = 0 \implies 7p^2 - 24p - 16 &= 0 \quad \textbf{[1 mark]}\end{aligned}$$

Factorising $(7 \times (-16) = -112;\ -28 \times 4 = -112$ and $-28 + 4 = -24)$:

$$\begin{aligned}7p^2 - 28p + 4p - 16 = 0 &\implies 7p(p - 4) + 4(p - 4) = 0\\ &\implies (p - 4)(7p + 4) = 0 \implies p = 4 \text{ or } p = -\frac{4}{7} \quad \textbf{[1 mark]}\end{aligned}$$

Now finding the equal roots for each value of p:

- When $p = 4$: $9x^2 - 30x + 25 = 0 \implies (3x - 5)^2 = 0 \implies x = \dfrac{5}{3}, \dfrac{5}{3}$.
- When $p = -\dfrac{4}{7}$: $\left(-\dfrac{8}{7} + 1\right)x^2 - (-4 + 2)x + (-4 - 3) = 0 \implies -\dfrac{1}{7}x^2 + 2x - 7 = 0$
  $\implies x^2 - 14x + 49 = 0 \implies (x - 7)^2 = 0 \implies x = 7,\ 7.$

Hence, $\boxed{p = 4 \text{ (roots: } \frac{5}{3}, \frac{5}{3}) \text{ and } p = -\frac{4}{7} \text{ (roots: } 7, 7)}$. **[1 mark]**

Q10. Case Study: Petroleum Industry & Crude Oil Production Economics

An oil refinery models its daily net profit $P(x)$ (in rupees) from processing x barrels of crude oil per day by the quadratic relation $P(x) = -10x^2 + 3500x - 66000$, where the plant's maximum daily processing capacity is 300 barrels ($0 \le x \le 300$). [4]

(i) Write and simplify the quadratic equation corresponding to the break-even condition $P(x) = 0$. **[1]**

(ii) Find the discriminant of this break-even quadratic equation and comment on the existence of real break-even production levels. **[1]**

(iii) Find the minimum daily production (in barrels) needed to break even within plant capacity. **OR** By completing the square, find the daily output x that maximises profit and the maximum daily profit. **[2]**

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

We know the break-even condition occurs when profit equals zero ($P(x) = 0$), and the vertex of a quadratic is found by completing the square:

Formula Used: Break-even: $P(x) = 0$, $\quad ax^2 + bx + c = a(x-h)^2 + k$

(i) Break-Even Equation: Setting $P(x) = 0$:

$$-10x^2 + 3500x - 66000 = 0 \implies \text{dividing by } (-10): \boxed{x^2 - 350x + 6600 = 0} \quad \textbf{[1 mark]}$$

(ii) Discriminant: For the reduced equation $x^2 - 350x + 6600 = 0$ ($a = 1, b = -350, c = 6600$):

$$D = (-350)^2 - 4(1)(6600) = 122500 - 26400 = \boxed{96100} = (310)^2 > 0$$

(Note: Using unreduced coefficients $a = -10, b = 3500, c = -66000$ gives $D_{\text{orig}} = 96{,}10{,}000 = 3100^2 > 0$.)

Since $D > 0$, two distinct real break-even points exist. **[1 mark]**

(iii) First Part (Minimum Break-Even Output): Solving $x^2 - 350x + 6600 = 0$ using $D = 310^2$:

$$x = \frac{350 \pm 310}{2} \implies x = \frac{40}{2} = 20 \quad \text{or} \quad x = \frac{660}{2} = 330$$

Since maximum daily plant capacity is 300 barrels, $x = 330 > 300$ exceeds capacity and is rejected.

Hence, minimum daily production to break even is $\boxed{20 \text{ barrels/day}}$. **[2 marks]**

OR (Alternative Part — Maximum Profit via Completing the Square): Rewriting $P(x)$:

$$\begin{aligned} P(x) &= -10(x^2 - 350x) - 66000 \\ &= -10\left[x^2 - 2(x)(175) + 175^2 - 175^2\right] - 66000 \\ &= -10(x-175)^2 + 10(30625) - 66000 = -10(x-175)^2 + 240250 \end{aligned}$$

Since $-10(x-175)^2 \le 0$ for all real x, $P(x)$ attains its maximum when $(x-175)^2 = 0$, i.e., at $x = 175 \le 300$.

Hence, $\boxed{\text{Optimal output} = 175 \text{ barrels/day}}$ and $\boxed{\text{Maximum profit} = \text{Rs. } 2{,}40{,}250}$. **[2 marks]**

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